Abraham Barreto | Group FISH

A02 "Image Processing Adventure Quest"

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2 September 2026

Individual Contribution Journal: The Royal Image Banquet

Researching BMP for the Royal Image Banquet changed how I think about image storage and file size. Our group selected the Kingdom of Image Formats and completed the Royal Banquet Menu challenge. My main responsibilities included researching BMP, contributing to the BMP section of the menu, and completing Slides 4 and 5 in Canva. I approached my work by studying BMP's technical features and then presenting them in a way that matched the creative banquet theme.

My research focused on BMP, or Bitmap, which we represented as the "BMP Burger." BMP was developed for Windows and commonly stores pixel data with little or no compression (Adobe). This approach can preserve image information, but it also creates very large files that are inefficient to store or share. BMP may support transparency in certain versions and applications, but its transparency support is less dependable than PNG. The burger theme represented the format's heavy file size, while the recommendation "Choose BMP when simple pixel storage matters more than file size" summarized its main use.

Using the PNG research completed by Viktoriya, I added and organized the information for the PNG Parfait slide in Canva. I also completed the BMP Burger slide using my own BMP research. Both slides followed the presentation's burgundy and gold design, food theme, and organized information panels. Each slide explained the format's best uses, compression type, strengths, trade-offs, and transparency support. I reviewed the poster and presentation to ensure that the information remained consistent with the research completed by the group. Comparing the formats showed me that JPEG prioritizes convenient sharing, PNG provides reliable transparency, BMP offers direct pixel storage, and TIFF focuses on preserving high-quality image information (Adobe).

In conclusion, before this project, I often selected JPEG or PNG without thinking carefully about the differences between formats. Researching BMP showed me how file formats are shaped by the technology and priorities of the time in which they were created. This also connects to computer vision because compression and pixel representation can affect preprocessing, memory use, and the image details available to a model. The project helped me understand that choosing an image format requires balancing quality, file size, transparency, compatibility, and purpose.

References

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